



KLE Technological
University
Creating Value
Leveraging Knowledge

**Department
of
Electronics and Communication Engineering**

**Mini Project Report
on
FAST MOBILE CHARGER**

By:

- | | |
|----------------------|--------------|
| 1. Achyut Bidkar | 01FE23BEC012 |
| 2. Bhavana A | 01FE23BEC087 |
| 3. Pradnya A Purohit | 01FE23BEC010 |
| 4. Pooja R K | 01FE23BEC080 |

Semester: V, 2025-2026

Under the Guidance of
Mr. Shamshuddin K

K.L.E SOCIETY'S
KLE Technological University,
HUBBALLI-580031
2025-26



**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

CERTIFICATE

This is to certify that project entitled **“Fast Mobile Charger”** is a Bonafide work carried out by the student team of **Achyut Bidkar(01FE23BEC012), Bhavana A(01FE23BEC087), Pradnya A Purohit(01FE23BEC010), Pooja R K(01FE23BEC080)**. The project report has been approved as it satisfies the requirements with respect to the Mini project work pre- scribed by the university curriculum for BE (V semester) in Department of Electronics and Communication Engineering of KLE Technological University for the academic year 2025- 26.

Mr. Shamshuddin K
Guide

Dr. Suneeta Budhihal
HoD

Dr. B. S. Anami
Registrar

External Viva:

Name of Examiners

Signature with date

- 1.
- 2.

ACKNOWLEDGMENT

The satisfaction that accompanies the successful completion of any task would be incomplete without acknowledging those who made it possible, as success is the result of hard work, determination, concentration, and dedication. I express my sincere gratitude to my project guide Mr. Shamshuddin K and project coordinator Mr. Sachidanand B. N. for their valuable guidance and constructive suggestions, which directed me toward the successful completion of this project. I am deeply thankful to the Head of the Department of Electronics and Communication Engineering, Dr. Suneeta Budihal, for her constant encouragement and kind cooperation throughout the project work. I also take this opportunity to express my heartfelt thanks to our respected Registrar, Dr.. B. S. Anami, for his continuous support and motivation, which greatly contributed to the successful completion of this project. Lastly, I extend my sincere thanks to all the faculty and staff members who directly or indirectly supported me during the completion of this project report.

ABSTRACT

This mini project presents the design, simulation, and hardware implementation of a regulated DC power supply circuit intended for fast and stable mobile phone charging applications. The proposed system converts an AC input supply into a smooth and regulated DC output using a combination of power electronic components, ensuring reliable voltage delivery suitable for low-power electronic devices. Emphasis is placed on voltage stability, ripple reduction, and practical feasibility using discrete components.

The circuit employs a step-down transformer to reduce the AC mains voltage to the required level, followed by a full-wave bridge rectifier for AC–DC conversion. A large electrolytic capacitor is used at the rectifier output to significantly reduce ripple content and improve the quality of the DC signal. Voltage regulation is achieved through a transistor-based regulator circuit controlled by a Zener diode reference, which maintains a nearly constant output voltage despite variations in input voltage and load conditions. Current-limiting resistors are incorporated to protect the circuit components and the connected load.

Prior to hardware implementation, the circuit was simulated to verify rectification efficiency, filtering performance, and regulated output characteristics. The simulated results provided insight into voltage levels at various stages of the circuit, enabling optimization of component values. The hardware prototype was constructed on a breadboard and tested using a digital multi-meter to measure output voltage and current. The experimental results demonstrate that the circuit delivers a stable DC output with minimal fluctuation, closely matching the simulated and theoretical values.

The developed power supply is simple in design, cost-effective, and reliable, making it suitable for educational purposes and basic charging applications. This project reinforces fundamental concepts of power electronics such as rectification, filtering, and linear voltage regulation, while also highlighting practical considerations involved in implementing real-world DC power supply systems.

Contents

1	Introduction	1
1.1	Motivation.....	1
1.2	Objectives.....	1
1.3	Problem statement.....	2
1.4	Organization of report.....	2
2	Electronic System design	3
2.1	Alternative solutions.....	4
2.2	Functional block diagram.....	5
3	Implementation detail	6
3.1	Specification and system design.....	6
3.2	Flowchart.....	7
4	Results and discussions	7
4.1	Simulation Analysis.....	8
4.2	Hardware Realization.....	8
5	Conclusion and Reflections	10
5.1	Conclusion.....	10
5.2	Future scope.....	10
5.2.1	Application in societal content.....	10
	References	13

List of Tables

2.1	Table.1 Components used.....	3
-----	------------------------------	---

List of Figures

1.1 Logo of KLE technological University.....	9
2.1 Components Interconnected on breadboard according to simulation schematic.....	3
2.2 Circuit of 12V-1A.....	4
2.3 Circuit of 5V-3A.....	4
2.4 Functional Block Diagram.....	5
2.2 Function block diagram of the fast mobile charger.....	5
3.1 Transmitter and Receiver Flow Diagram.....	6
4.1 Final results of Implementation.....	8

Chapter 1

Introduction

The widespread use of mobile devices has increased the need for dependable and safe charging systems. This project focuses on the development of a low-cost mobile charger based on a regulated DC power supply that delivers a stable output with reduced ripple. The straightforward linear design makes the system well suited for educational and laboratory applications.

1.1 Motivation

The increasing dependence on mobile phones and portable electronic devices in everyday life has created a strong need for reliable, efficient, and safe charging systems. Mobile chargers play a critical role in ensuring uninterrupted device operation, and any instability in charging voltage or current can negatively affect battery life, device performance, and user safety. This growing demand for dependable charging solutions motivated the selection of a regulated DC power supply-based mobile charger as the focus of this mini project.

Commercial fast chargers available in the market often use complex switching circuits and proprietary charging protocols, which increase cost and reduce transparency in understanding their operation. For engineering students, such designs offer limited insight into fundamental power electronics concepts. This project is motivated by the need to develop a simple, low-cost charging system using discrete electronic components that clearly demonstrate the core principles of rectification, filtering, and voltage regulation.

1.2 Objectives

The objective of this project is to design and implement a low-cost mobile charger using a regulated DC power supply that provides stable voltage and sufficient current for safe charging. The system applies fundamental power electronics concepts such as voltage step-down, rectification, filtering, and regulation. A Zener diode-based regulation stage ensures consistent output under varying input and load conditions. The charger's performance is evaluated through simulation and hardware testing to verify voltage stability, ripple reduction, and reliable operation.

1.3 Problem statement

“Design and implement a 5 V, 1 A regulated DC linear power supply using discrete components. The system should convert AC mains voltage into a stable 5 V DC output suitable for low-power electronic applications. The design must be simulated and practically implemented, and differences between theoretical and practical results must be analyzed.”

The objective of this mini project is to design and implement a low-cost fast mobile charger capable of delivering stable and regulated DC output suitable for charging mobile electronic devices. Mobile chargers are required to provide consistent voltage and sufficient current to ensure safe and efficient charging while protecting the connected device from voltage fluctuations, current surges, and excessive ripple.

Many commercially available charging solutions rely on complex switching circuits, which increase cost and design complexity and are not always suitable for educational or small-scale applications. This project aims to address these challenges by developing a simple yet reliable linear Regulated power supply using discrete electronic components. The system is designed to convert The available 230V AC mains supply into a low-voltage DC output of approximately 5V, which Meets the basic charging requirements of mobile devices. Emphasis is placed on voltage regulation, ripple reduction, and current handling capability to ensure stable operation under varying load conditions.

The proposed charger is intended to be economical, easy to construct, and suitable for Laboratory implementation while demonstrating core concepts of power electronics such as rectification, filtering, and regulation. The design also considers component availability, safety, and performance, making it appropriate for academic mini project applications.

1.4 Organization of the report

Introduction

The growing use of mobile devices necessitates safe and reliable charging solutions. This project designs a low-cost regulated DC power supply-based mobile charger with stable voltage and minimal ripple. The simple linear design is suitable for educational and laboratory use.

Electronic system design

The proposed work designs and implements a low-cost regulated DC power supply-based fast mobile charger emphasizing simplicity and reliability. AC mains are converted into a stable, low-ripple DC output using transformation, rectification, filtering, and voltage regulation with discrete components. The hardware implementation validates the design and demonstrates safe mobile charging, making it suitable for educational and low-cost applications.

Implementation details

The system specifications and design of the proposed fast mobile charger, including hardware and software components. The design uses discrete electronic components for stable voltage regulation and reliable operation, with implementation carried out on a breadboard. Circuit simulation using Proteus and Multi-sim is performed prior to hardware testing, and a flow chart illustrates the operational sequence of the charging process.

Results and Discussions

This chapter presents the simulation and hardware results of the regulated DC power supply for fast mobile charging. Simulation and experimental testing verify proper voltage step-down, rectification, filtering, and regulation under both no-load and load conditions. The results confirm stable DC output with reduced ripple, validating the effectiveness of the proposed design for mobile charging applications.

Conclusion and reflections

The project successfully demonstrates the design and implementation of a low-cost regulated DC power supply for fast mobile charging, achieving stable voltage output and reliable performance. Simulation and experimental testing confirm proper voltage step-down, rectification, filtering, and regulation under both no-load and load conditions. The results show a stable DC output with reduced ripple, validating the proposed design for mobile charging applications.

Chapter 2

Electronic System design

The proposed work focuses on the design and implementation of a low-cost fast mobile charging system using a regulated DC power supply. The system converts the available AC mains supply into a stable, low-ripple DC output suitable for mobile devices. The design emphasizes simplicity, reliability, and practical feasibility using discrete electronic components. The proposed model ensures regulated voltage delivery, adequate current capability, and safe operation for charging applications.

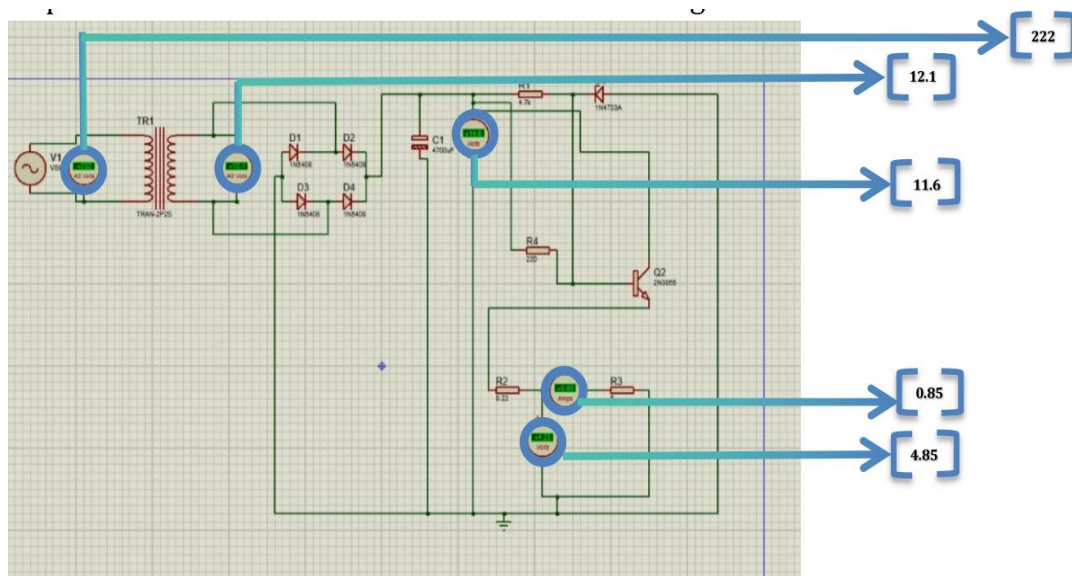


Fig.2.1 simulation schematic

In Fig.2.1 illustrates the proposed regulated power supply model for fast mobile charging applications. The 230V AC mains supply is first stepped down using a transformer and then converted into DC using a full-wave bridge rectifier. A filter capacitor smooths the rectified output by reducing ripple voltage. The voltage regulation stage, consisting of a Zener diode and power transistor, ensures a stable DC output of approximately 5V. The regulated output is monitored using a digital multimeter and is finally used to charge a mobile device safely and efficiently.

Table.2.1 Components used

Component	Specification	Quantity
Transformer	230V AC to 12V AC	1
Diode	1N5408	4
Zener Diode	1N4733A (5.1V)	1
Transistor	2N3055	1
Resistor	4.7 k Ω , 220 Ω , 5 Ω , 0.22 Ω	1 each
Capacitor	4700 μ F / 25V	1

2.1 Alternative Solutions

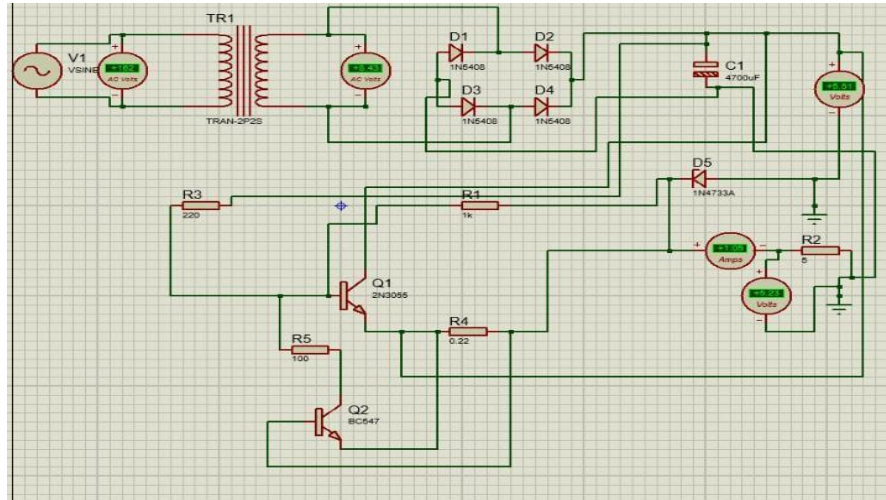


Fig.2.2 Circuit of 12V-1A

This alternative circuit implements a regulated DC power supply capable of delivering approximately 12 V at 1 A. The AC input is stepped down using a transformer and converted into DC through a full-wave bridge rectifier, after which a filter capacitor smooths the output by reducing ripple. Voltage regulation is achieved using a Zener diode in combination with a power transistor (2N3055), allowing the circuit to supply higher load current while maintaining a stable output voltage. An additional transistor stage provides current control and protection under overload conditions. This design demonstrates a higher-voltage, higher-current regulated supply suitable for general-purpose electronic applications and experimental use.

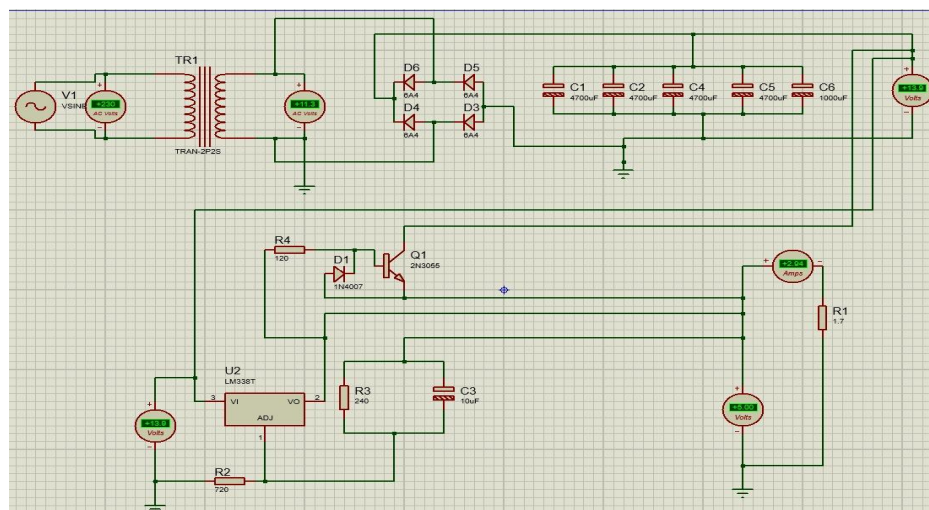


Fig2.3. Circuit of 5V-3A

This circuit represents a high-current regulated DC power supply, designed to convert 230 V AC mains into a stable low-voltage DC output suitable for heavy loads. The transformer (TR1) steps down the mains voltage to around 11.3 V AC, which is then fed to a bridge rectifier (D3–D6) to convert AC into pulsating DC.

To reduce ripple and improve smoothing, multiple large electrolytic capacitors (C1–C6) are connected in parallel, providing a high total capacitance that ensures a steady DC level at the rectifier output. This unregulated DC is then supplied to an LM338 adjustable voltage regulator, which is configured using resistors R2 and R3 to set the desired output voltage. Since the LM338 alone has current limitations, a power transistor (2N3055) is added as a pass transistor to boost the output current capability. The diode D1 and resistor R4 protect the regulator and help in proper current sharing, while capacitor C3 improves stability and transient response. Finally, the load resistor R1 demonstrates the output, showing a regulated voltage of around 5 V with a current close to 3 A, making this circuit suitable for applications requiring a reliable, high-current DC supply.

2.2 Functional block diagram

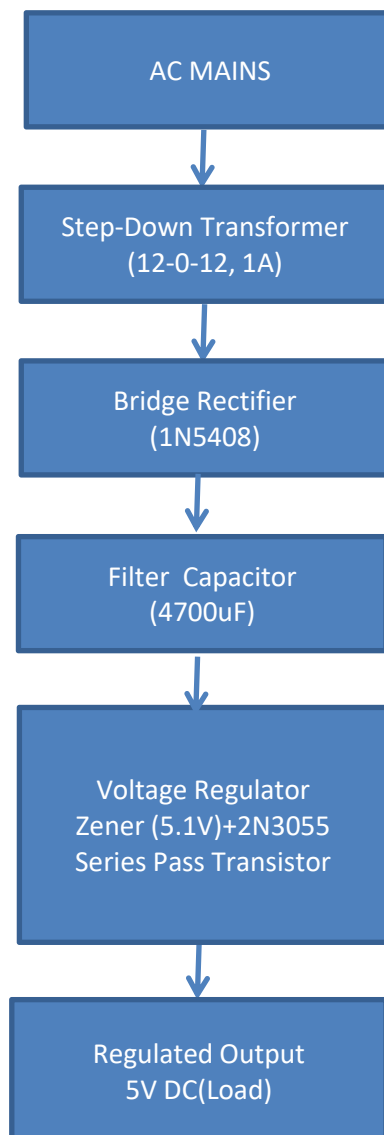


Fig.2.4. Functional Block Diagram of working of a regulated 5 V DC

This flowchart explains the working of a regulated 5 V DC power supply step by step. The AC mains supply is first applied to a step-down transformer (12-0-12 V, 1 A), which reduces the high mains voltage to a safer low-voltage AC suitable for electronic circuits.

This low-voltage AC is then fed to a bridge rectifier (using 1N5408 diodes), where the AC is converted into pulsating DC.

To smooth out the ripples present in the rectified output, a filter capacitor of 4700 μF is used, which stores charge and provides a nearly steady DC voltage. The smoothed DC is then applied to the voltage regulation stage, where a 5.1 V Zener diode provides a stable reference voltage and a 2N3055 series pass transistor increases the current handling capability while maintaining regulation. Finally, this stage delivers a regulated 5 V DC output to the load, ensuring constant voltage even with variations in input supply or load current, making the circuit suitable for powering sensitive electronic devices.

Chapter 3

Implementation details

The proposed fast mobile charger is designed using discrete hardware and supporting software tools to ensure stable voltage regulation and reliable operation. Circuit simulation is carried out using Proteus and Multi-sim before breadboard implementation, and a flowchart is used to represent the operational sequence of the charging process.

3.1 Specifications and system Design of Hardware and software specifications:

The hardware components used in the implementation of the fast mobile charger are selected to ensure stable voltage regulation, sufficient current handling capability, and reliable performance. The complete system is constructed using discrete electronic components and tested on a breadboard.

3.1.1 Major Hardware Components

- AC Power Supply: 230 V AC, 50 Hz mains supply
- Step-Down Transformer: 230 V AC to 12 V AC, single phase
- Rectifier: Full-wave bridge rectifier using silicon diodes (1N4007 / 1N5408)
- Filter Capacitor: Electrolytic capacitor (typically 2200 μ F – 4700 μ F, 25 V)
- Voltage Regulator: Zener diode based voltage regulation circuit
- Power Transistor: NPN power transistor (e.g., 2N3055 / TIP series)
- Resistors: Current-limiting and biasing resistors of appropriate ratings
- Load: Mobile phone / USB charging port
- Measuring Instruments: Digital multimeter for voltage and current measurement
- Prototype Platform: Breadboard and connecting wires

3.1.2 Software Specifications

Software tools are used for circuit design, simulation, and documentation of the proposed system. These tools help in verifying the functionality of the circuit before hardware implementation.

Software Tools Used

- Circuit Simulation Software: Proteus / Multisim (for simulation and verification)
- Design Tool: Proteus schematic editor
- Documentation Software: Microsoft Word / WPS Office for report preparation
- Operating System: Windows 10 or higher
- Simulation Parameters: Real-time voltage and current analysis, DC operating point and load variation testing

3.2 Flowchart

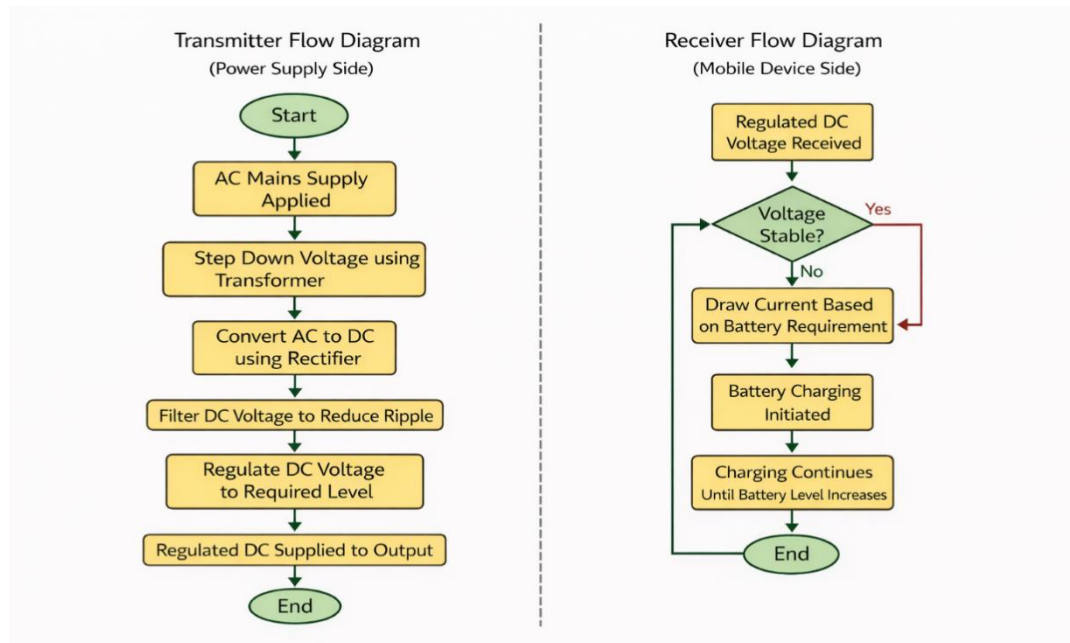


Fig. 2.4 Transmitter and Receiver Flow Diagram

The flowchart provides a structured representation of how electrical power is processed and delivered from the source to the mobile device in a controlled and systematic manner. The transmitter side emphasizes the internal functioning of the power supply unit, where electrical energy undergoes multiple stages of conditioning to ensure suitability for charging applications. Each block in the transmitter section highlights a specific function, such as voltage transformation, conversion, smoothing, and regulation, ensuring that the raw input power is gradually refined into a usable form. This sequence ensures electrical isolation, minimizes fluctuations, and prepares the output for safe interfacing with sensitive electronic devices. The step-by-step flow also reflects the logical dependency between stages, where the proper functioning of one stage is essential for the next, thereby improving overall system reliability.

On the receiver side, the flowchart focuses on how the regulated DC output is utilized by the mobile device during the charging process. Instead of directly accepting power, the device evaluates voltage stability to ensure safe operating conditions before drawing current. This conditional operation prevents improper charging under unstable voltage conditions and helps protect the battery from stress or damage. Once stability is confirmed, the charging current is drawn dynamically based on the battery's requirements, allowing controlled energy absorption. The charging process continues progressively until the battery level increases, after which the operation concludes safely. This receiver-side logic highlights the importance of controlled current flow, voltage verification, and gradual energy transfer, ensuring both device safety and efficient charging performance.

Chapter 4

Results and discussions

This chapter discusses the simulation and hardware results of the regulated DC power supply developed for fast mobile charging. Both simulation and experimental testing confirm effective voltage conversion, regulation, and ripple reduction under no-load and load conditions, validating the reliability of the proposed design.

4.1 Simulation Analysis

This describes the simulation procedure, experimental results, project outcomes, result snapshots, and graphical representation of the proposed regulated DC power supply for fast mobile charging. The performance of the system is validated through circuit simulation and hardware testing.

4.1.1 Simulation Procedure

The simulation of the proposed fast mobile charger was carried out to verify the functionality and performance of the circuit before hardware implementation. The circuit was designed using electronic simulation software, where individual components such as the transformer, rectifier diodes, filter capacitor, Zener diode, and power transistor were configured according to the design specifications.

Initially, the AC input source was set to represent the mains supply conditions. The step-down transformer was modeled to reduce the input voltage to the desired secondary voltage. The full-wave bridge rectifier was then connected to convert the AC voltage into pulsating DC. A filter capacitor was added at the rectifier output to minimize ripple voltage. Finally, the voltage regulation stage using a Zener diode and transistor was incorporated to obtain a stable DC output.

Simulation instruments such as virtual voltmeters and ammeters were placed at appropriate points in the circuit to observe voltage and current levels. The circuit was simulated under no-load and load conditions to analyze voltage regulation, current capability, and overall system behavior.

4.2 Hardware Realization

4.2.1 Simulation Results

The simulation results confirm the proper operation of each stage of the circuit. The observed values closely match the theoretical design expectations.

- Input Voltage: 222 V AC
- Transformer Secondary Voltage: 12.1 V AC
- Rectified DC Voltage: 11.6 V DC
- Regulated Output Voltage: 4.85 V DC
- Output Current: 0.85 A

The regulated output voltage remains stable even with variations in input voltage and load, demonstrating effective voltage regulation.

4.2.2 Hardware Results (Project Outcomes)

After successful simulation, the circuit was implemented on a breadboard using discrete components. The hardware testing was performed using a digital multimeter to measure output voltage and current.

The experimental results confirm that the hardware prototype produces a stable DC output suitable for mobile charging. The charger successfully delivers approximately 5V DC with sufficient current capability to charge a mobile device. The practical output values closely align with the simulated results, validating the design methodology.

4.2.3 Project Outcomes:

- Successful conversion of AC mains to regulated DC output
- Stable output voltage suitable for mobile charging
- Reduced ripple voltage due to effective filtering
- Reliable operation under load conditions
- Low-cost and simple implementation using discrete components

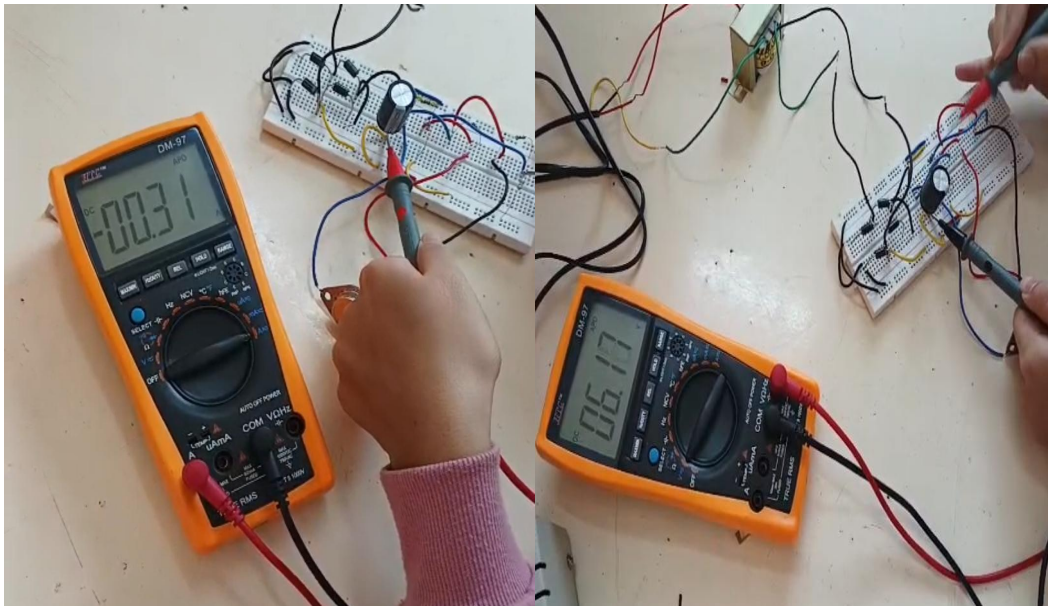


Fig.4.1 Final results of Implementation

This image shows the experimental setup of the fast mobile charger circuit implemented on a breadboard. A digital multi-meter is connected at the output terminals to measure the regulated DC voltage produced by the charger. The displayed reading confirms that the circuit delivers a stable low-voltage DC output suitable for mobile charging. The presence of rectifier diodes, filter capacitor, and regulation components on the breadboard verifies correct hardware assembly. This setup demonstrates successful voltage regulation and validates the effectiveness of the designed power supply.

This image illustrates the testing process of the hardware prototype under load conditions. The digital multi-meter probes are connected to different points in the circuit to observe voltage and current behavior during operation. The step-down transformer, rectification stage, and filtering components are clearly visible, indicating practical implementation of the simulated design. The measured values closely align with simulation results, confirming stable performance. This testing confirms that the charger operates reliably and safely during real-time mobile charging.

Chapter 5

Conclusion and Reflections

The project successfully implements a low-cost regulated DC power supply for fast mobile charging with stable and reliable performance. Simulation and hardware results confirm effective voltage conversion, regulation, and ripple reduction under both no-load and load conditions, validating the proposed design for charging applications.

5.1 Conclusion

In this mini project, a low-cost fast mobile charger based on a regulated DC power supply has been successfully designed, simulated, and implemented. The proposed system effectively converts the 230V AC mains supply into a stable DC output suitable for mobile charging applications. The design incorporates essential power electronics stages such as step-down transformation, rectification, filtering, and voltage regulation.

Simulation results verified the functionality of the circuit and provided insights into voltage levels and current flow at various stages. The hardware implementation closely matched the simulated results, demonstrating the accuracy and reliability of the design. The charger delivered a regulated output voltage of approximately 5V DC with sufficient current capacity, enabling safe and effective mobile charging.

The project highlights the importance of voltage regulation and ripple reduction in charging applications. By using a Zener diode and power transistor-based regulation scheme, the system maintained stable output voltage under varying input and load conditions. The simplicity of the design and use of readily available components make it suitable for educational purposes and low-power applications.

Although the charger exhibits moderate efficiency due to the use of linear regulation, its advantages in terms of simplicity, reliability, and ease of implementation outweigh the limitations for academic and experimental use. The project successfully bridges theoretical knowledge and practical implementation, reinforcing fundamental concepts of power electronics.

Overall, the proposed fast mobile charger fulfills the objectives of providing a regulated, reliable, and economical charging solution. The insights gained from this project form a strong foundation for further advancements in efficient power supply and charging system design.

5.2 Future scope

5.2.1 Application in the societal context

Although the proposed fast mobile charger successfully delivers a regulated DC output suitable for basic charging applications, there are several areas where the system can be further improved and extended. These enhancements can increase efficiency, safety, functionality, and applicability in real-world scenarios.

One significant area of future work is improving the efficiency of the power supply. The present design uses linear voltage regulation, which results in power loss in the form of heat. Future versions of the charger can incorporate switched-mode power supply (SMPS) techniques such as buck converters. These converters offer higher efficiency, reduced heat dissipation, and compact size, making the charger more suitable for commercial applications.

Another potential improvement is the integration of intelligent charging control. By using microcontrollers or dedicated charging ICs, features such as automatic current control, battery status monitoring, and adaptive charging based on battery condition can be implemented. This would enhance charging speed while ensuring battery safety and longevity.

Advanced protection mechanisms can also be added as part of future work. Protection features such as short-circuit protection, over-current protection, over-voltage protection, and thermal shutdown would improve the safety and reliability of the charger. These features are especially important for prolonged usage and high-current operation.

The charger design can further be enhanced by replacing the bulky step-down transformer with high-frequency transformer technology used in SMPS designs. This would significantly reduce the size and weight of the charger, making it more portable and suitable for consumer use.

Another scope of future work is the inclusion of USB standard charging interfaces and fast-charging protocols. Supporting widely used standards such as USB-based charging and fast-charge modes would allow compatibility with a broader range of devices and improve charging performance.

The system can also be expanded to support renewable energy sources. By integrating solar panels or battery backup systems, the charger can be adapted for off-grid or emergency charging applications. This would increase its usability in rural or remote areas.

Bibliography

- [1] M. H. Rashid, *Power Electronics: Circuits, Devices and Applications*, 4th ed., New Delhi, India: Pearson Education, 2014, pp. 35–78.
- [2] P. S. Bimbhra, *Power Electronics*, 6th ed., New Delhi, India: Khanna Publishers, 2015, pp. 112–145.
- [3] D. P. Kothari and I. J. Nagrath, *Electric Machines*, 4th ed., New Delhi, India: McGraw-Hill Education, 2011, pp. 521–548.
- [4] R. L. Boylestad and L. Nashelsky, *Electronic Devices and Circuit Theory*, 11th ed., Pearson Education, 2013, pp. 742–789.
- [5] A. P. Malvino and D. Bates, *Electronic Principles*, 8th ed., New York, USA: McGraw-Hill, 2015, pp. 601–635.
- [6] R. A. Gayakwad, *Op-Amps and Linear Integrated Circuits*, 4th ed., New Delhi, India: Pearson Education, 2014, pp. 383–402.
- [7] J. Millman and C. C. Halkias, *Integrated Electronics: Analog and Digital Circuits and Systems*, New York, USA: McGraw-Hill, 2010, pp. 457–490.
- [8] Texas Instruments, *Voltage Regulator Handbook*, Application Report, 2018, pp. 1–28.
- [9] ON Semiconductor, *Zener Diode Theory and Applications*, Application Note, 2017, pp. 5–19.
- [10] A. K. Sawhney, *A Course in Electrical and Electronic Measurements and Instrumentation*, New Delhi, India: Dhanpat Rai & Co., 2016, pp. 302–325.
- [11] S. B. Kulkarni, “Design and implementation of regulated DC power supply,” *International Journal of Engineering Research and Technology (IJERT)*, vol. 4, no. 6, pp. 210–214, 2015.
- [12] R. Singh and A. Verma, “Design of low-cost mobile charger using linear regulation,” *International Journal of Electronics and Communication Engineering*, vol. 7, no. 2, pp. 85–90, 2016.
- [13] P. Kumar, “Analysis of rectifier and filter circuits for DC power supplies,” *International Journal of Electrical Engineering*, vol. 9, no. 3, pp. 45–52, 2017.
- [14] B. W. Williams, *Power Electronics: Devices, Drivers and Applications*, London, UK: Macmillan Press, 2012, pp. 158–196.
- [15] IEEE Standards Association, *IEEE Guide for Power Supply Design*, IEEE Std 101, 2019, pp. 12–13.